

- 7.2 a) Figure 7.3 describes the electromagnetic spectrum by wavelength and frequency. Wavelength increases from left (10^{-2} nm) to right (10^{12} nm). The trend in increasing wavelength is: **x-ray < ultraviolet < visible < infrared < microwave < radio waves**.
- b) Frequency is inversely proportional to wavelength according to equation 7.1, so frequency has the opposite trend: **radio < microwave < infrared < visible < ultraviolet < x-ray**.
- c) Energy is directly proportional to frequency according to equation 7.2. Therefore, the trend in increasing energy matches the trend in increasing frequency: **radio < microwave < infrared < visible < ultraviolet < x-ray**. High-energy electromagnetic radiation disrupts cell function. It makes sense that you want to limit exposure to ultraviolet and x-ray radiation.

7.7 Plan: Wavelength is related to frequency through the equation $c = \lambda\nu$. Recall that a Hz is a reciprocal second, or $1/s = s^{-1}$. Assume that the number “960” has three significant figures.

Solution:

$$c = \lambda\nu$$

$$\lambda(\text{m}) = c/\nu = \frac{3.00 \times 10^8 \text{ m/s}}{(960. \text{ kHz}) \left(\frac{10^3 \text{ Hz}}{1 \text{ kHz}} \right) \left(\frac{s^{-1}}{\text{Hz}} \right)} = 312.5 = \mathbf{312 \text{ m}}$$

$$\lambda(\text{nm}) = c/\nu = \frac{3.00 \times 10^8 \text{ m/s}}{(960. \text{ kHz}) \left(\frac{10^3 \text{ Hz}}{1 \text{ kHz}} \right) \left(\frac{s^{-1}}{\text{Hz}} \right)} \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 3.125 \times 10^{11} = \mathbf{3.12 \times 10^{11} \text{ nm}}$$

$$\lambda(\text{\AA}) = c/\nu = \frac{3.00 \times 10^8 \text{ m/s}}{(960. \text{ kHz}) \left(\frac{10^3 \text{ Hz}}{1 \text{ kHz}} \right) \left(\frac{s^{-1}}{\text{Hz}} \right)} \left(\frac{0.01 \text{\AA}}{10^{-12} \text{ m}} \right) = 3.125 \times 10^{12} = \mathbf{3.12 \times 10^{12} \text{\AA}}$$

7.9 Plan: Frequency is related to energy through the equation $E = h\nu$. Note that $1 \text{ Hz} = 1 \text{ s}^{-1}$.

Solution:

$$E = (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) (3.6 \times 10^{10} \text{ s}^{-1}) = 2.385 \times 10^{-23} = \mathbf{2.4 \times 10^{-23} \text{ J}}$$

7.11 Since energy is directly proportional to frequency ($E = h\nu$) and frequency and wavelength are inversely related ($\nu = c/\lambda$), it follows that energy is inversely related to wavelength. As wavelength decreases, energy increases. In terms of increasing energy the order is **red < yellow < blue**.

7.13 Plan: Frequency and wavelength can be calculated using the speed of light: $c = \lambda\nu$.

Solution:

$$\lambda(\text{nm}) = c/\nu = \frac{2.99792 \times 10^8 \text{ m/s}}{(22.235 \text{ GHz}) \left(\frac{10^9 \text{ Hz}}{1 \text{ GHz}} \right) \left(\frac{s^{-1}}{\text{Hz}} \right)} \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 1.3482887 \times 10^7 = \mathbf{1.3483 \times 10^7 \text{ nm}}$$

$$\lambda(\text{\AA}) = c/\nu = \frac{2.99792 \times 10^8 \text{ m/s}}{(22.235 \text{ GHz}) \left(\frac{10^9 \text{ Hz}}{1 \text{ GHz}} \right) \left(\frac{s^{-1}}{\text{Hz}} \right)} \left(\frac{0.01 \text{\AA}}{10^{-12} \text{ m}} \right) = 1.3482887 \times 10^8 = \mathbf{1.3483 \times 10^8 \text{\AA}}$$

7.16 Plan: a) The least energetic photon has the longest wavelength (242 nm). b) The most energetic photon has the shortest wavelength (2200 \AA).

Solution:

$$\text{a) } \nu = c/\lambda = \frac{3.00 \times 10^8 \text{ m/s}}{242 \text{ nm}} \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 1.239669 \times 10^{15} = \mathbf{1.24 \times 10^{15} \text{ s}^{-1}}$$

$$E = hc/\lambda = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s}) (3.00 \times 10^8 \text{ m/s})}{242 \text{ nm}} \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 8.2140 \times 10^{-19} = \mathbf{8.21 \times 10^{-19} \text{ J}}$$

$$\text{b) } \nu = c/\lambda = \frac{3.00 \times 10^8 \text{ m/s}}{2200 \text{ \AA}} \left(\frac{0.01 \text{\AA}}{10^{-12} \text{ m}} \right) = 1.3636 \times 10^{15} = \mathbf{1.4 \times 10^{15} \text{ s}^{-1}}$$

$$E = hc/\lambda = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s}) (3.00 \times 10^8 \text{ m/s})}{2200 \text{ \AA}} \left(\frac{0.01 \text{\AA}}{10^{-12} \text{ m}} \right) = 9.03545 \times 10^{-19} = \mathbf{9.0 \times 10^{-19} \text{ J}}$$

- 7.20 The quantum number n is related to the energy level of the electron. An electron *absorbs* energy to change from lower energy (low n) to higher energy (high n) giving an absorption spectrum. An electron *emits* energy as it drops from a higher energy level to a lower one giving an emission spectrum.
a) **absorption** b) **emission** c) **emission** d) **absorption**
- 7.22 The Bohr model has successfully predicted the line spectra for the H atom and Be^{3+} ion since both are one electron species. The energies could be predicted from $E_n = \frac{-(Z^2)(2.18 \times 10^{-18} \text{ J})}{n^2}$ where Z is the atomic number for the atom or ion. The line spectra for H would not match the line spectra for Be^{3+} since the H nucleus contains one proton while the Be^{3+} nucleus contains 4 protons (the Z values in the equation do not match), thus the force of attraction of the nucleus for the electron would be greater in the beryllium ion than in the hydrogen atom. This means that the pattern of lines would be similar, but at different wavelengths.
- 7.23 Plan: Calculate wavelength by substituting the given values into equation 7.3, where $n_1 = 2$ and $n_2 = 5$ because $n_2 > n_1$. Although more significant figures could be used, five significant figures are adequate for this calculation.
Solution:
- $$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad R = 1.096776 \times 10^7 \text{ m}^{-1}$$
- $$n_1 = 2 \quad n_2 = 5$$
- $$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = (1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{2^2} - \frac{1}{5^2} \right) = 2303229.6 \text{ m}^{-1} \text{ (unrounded)}$$
- $$\lambda \text{ (nm)} = \left(\frac{1}{2303229.6 \text{ m}^{-1}} \right) \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 434.1729544 = \mathbf{434.17 \text{ nm}}$$
- 7.25 Plan: The Rydberg equation is needed. For the infrared series of the H atom, n_1 equals 3. The least energetic spectral line in this series would represent an electron moving from the next highest energy level, $n_2 = 4$. Although more significant figures could be used, five significant figures are adequate for this calculation.
Solution:
- $$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = (1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = 533155 \text{ m}^{-1} \text{ (unrounded)}$$
- $$\lambda \text{ (nm)} = \left(\frac{1}{533155 \text{ m}^{-1}} \right) \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right) = 1875.627 = \mathbf{1875.6 \text{ nm}}$$
- Check: Checking this wavelength with Figure 7.9, we find the line in the infrared series with the greatest wavelength occurs at approximately 1850 nm.

7.31 Plan: Use the Rydberg equation. A combination of $E = hc/\lambda$, and equation 7.4, would also work.

Solution:

$$\lambda = (97.20 \text{ nm}) \times (10^{-9} \text{ m/1 nm}) = 9.720 \times 10^{-8} \text{ m}$$

$$1/\lambda = (1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$1/(9.720 \times 10^{-8} \text{ m}) = (1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{1^2} - \frac{1}{n_2^2} \right)$$

$$0.9380 = \left(\frac{1}{1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{n_2^2} = 1 - 0.9380 = 0.0620$$

$$n_2^2 = 16.1$$

$$n_2 = 4$$

7.39 Plan: Part a) uses the de Broglie equation, and part b) uses Heisenberg's relationship.

Solution:

$$\begin{aligned} \text{a) } \lambda = h/mv &= \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}} \right) \left(\frac{2.205 \text{ lb}}{1 \text{ kg}} \right) \left(\frac{0.62 \text{ mi}}{1 \text{ km}} \right) \left(\frac{1 \text{ km}}{10^3 \text{ m}} \right) \left(\frac{3600 \text{ s}}{1 \text{ h}} \right)}{(220 \text{ lb}) \left(19.6 \frac{\text{mi}}{\text{h}} \right)} \\ &= 7.562675 \times 10^{-37} = \mathbf{7.6 \times 10^{-37} \text{ m}} \end{aligned}$$

7.43 Plan: The de Broglie wavelength equation will give the mass equivalent of a photon with known wavelength and velocity. The term "mass-equivalent" is used instead of "mass of photon" because photons are quanta of electromagnetic energy that have no mass. A light photon's velocity is the speed of light, $3.00 \times 10^8 \text{ m/s}$.

Solution:

$$\lambda = h/mv$$

$$m = h/\lambda v = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}} \right) \left(\frac{1 \text{ nm}}{10^{-9} \text{ m}} \right)}{(589 \text{ nm}) (3.00 \times 10^8 \text{ m/s})} = 3.7498 \times 10^{-36} = \mathbf{3.75 \times 10^{-36} \text{ kg/photon}}$$

7.47 A peak in the radial probability distribution at a certain distance means that the total probability of finding the electron is greatest within a thin spherical volume having a radius very close to that distance. Since principal quantum number (n) correlates with distance from the nucleus, the peak for $n = 2$ would occur at a greater distance from the nucleus than $0.529 \text{ \AA}$. Thus, the probability of finding an electron at $0.529 \text{ \AA}$ is much greater for the $1s$ orbital than for the $2s$.

7.48 a) Principal quantum number, n , relates to the size of the orbital. More specifically, it relates to the distance from the nucleus at which the probability of finding an electron is greatest. This distance is determined by the energy of the electron.

b) Angular momentum quantum number, l , relates to the shape of the orbital. It is also called the azimuthal quantum number.

c) Magnetic quantum number, m_l , relates to the orientation of the orbital in space in three-dimensional space.

7.49 a) **one** b) **five** c) **three** d) **nine**

a) There is only a single s orbital in any shell.

b) All d -orbitals consists of sets of five ($m_l = -2, -1, 0, +1, +2$).

c) All p -orbitals consists of sets of three ($m_l = -1, 0, +1$).

d) If $n = 3$, then there is a $3s$ (1 orbital), a $3p$ (3 orbitals), and a $3d$ (5 orbitals) giving $1 + 3 + 5 = 9$.

7.51 Magnetic quantum numbers can have integer values from $-l$ to $+l$.

a) m_l : -2, -1, 0, +1, +2

b) m_l : 0 (if $n = 1$, then $l = 0$)

c) m_l : -3, -2, -1, 0, +1, +2, +3

7.57 a) For the $5s$ subshell, $n = 5$ and $l = 0$. Since $m_l = 0$, there is one orbital.

b) For the $3p$ subshell, $n = 3$ and $l = 1$. Since $m_l = -1, 0, +1$, there are three orbitals.

c) For the $4f$ subshell, $n = 4$ and $l = 3$. Since $m_l = -3, -2, -1, 0, +1, +2, +3$, there are seven orbitals.

7.62 a) The mass of the electron is 9.1094×10^{-31} kg.

$$E = -\frac{h^2}{8\pi^2 m_e a_0^2 n^2} = -\frac{h^2}{8\pi^2 m_e a_0^2} \left(\frac{1}{n^2} \right)$$

$$= -\frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^2}{8\pi^2 (9.1094 \times 10^{-31} \text{ kg}) (52.92 \times 10^{-12} \text{ m})^2} \left(\frac{\text{kg} \cdot \text{m}^2/\text{s}^2}{\text{J}} \right) \left(\frac{1}{n^2} \right)$$

$$= -(2.17963 \times 10^{-18} \text{ J}) \left(\frac{1}{n^2} \right) = -(2.180 \times 10^{-18} \text{ J}) \left(\frac{1}{n^2} \right)$$

This is identical with the result from Bohr's theory. For the H atom, $Z = 1$ and Bohr's constant $= -2.18 \times 10^{-18}$ J. For the hydrogen atom, derivation using classical principles or quantum-mechanical principles yields the same constant.

b) The $n = 3$ energy level is higher in energy than the $n = 2$ level. Because the zero point of the atom's energy is defined as an electron's infinite distance from the nucleus, a larger negative number describes a lower energy level. Although this may be confusing, it makes sense that an energy *change* would be a positive number.

$$\Delta E = -(2.180 \times 10^{-18} \text{ J}) \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = -3.027778 \times 10^{-19} = \mathbf{3.028 \times 10^{-19} \text{ J}}$$

c) Find the frequency that corresponds to this energy using $\nu = E/h$, and then calculate wavelength using $\lambda = c/\nu$.

$$\lambda (\text{m}) = hc/E = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) (2.9979 \times 10^8 \text{ m/s})}{(3.027778 \times 10^{-19} \text{ J})} = 6.56061 \times 10^{-7} = \mathbf{6.561 \times 10^{-13} \text{ m}}$$

$$= 6.562 \times 10^{-7} \text{ m} = 656.2 \text{ nm}$$

This is the wavelength for the observed red line in the hydrogen spectrum.

7.75 a) The l value must be at least 1 for m_l to be -1, but cannot be greater than $n - 1 = 2$. Increase the l value to 1 or 2 to create an allowable combination.

b) The l value must be at least 1 for m_l to be +1, but cannot be greater than $n - 1 = 2$. Decrease the l value to 1 or 2 to create an allowable combination.

c) The l value must be at least 3 for m_l to be +3, but cannot be greater than $n - 1 = 6$. Increase the l value to 3, 4, 5, or 6 to create an allowable combination.

d) The l value must be at least 2 for m_l to be -2, but cannot be greater than $n - 1 = 3$. Increase the l value to 2 or 3 to create an allowable combination.